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Environment: The First Thing to Look at in Conceptual Design

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Abstract This paper aims to addresses three questions related to conceptual design: 1) Why is environment important? 2) What is environment? and 3) How to analyze environment? Unlike product-, function (affordance)-, or user-centered methodologies, it is argued that the known environment of a product to be designed must be the first thing to be looked at in a conceptual design. The environment can be categorized into three types: the natural, the built and the human. A complete set of environment for a design is implied in the lifecycle of a product to be designed, which can be further defined by lifecycle activities in terms of time. The analysis of environment consists of three activities: 1) determine decision directions by asking design questions based on a design problem statement; 2) clarify the design problem statement by answering questions following the decision direction; and 3) formulate an environment system from a design problem statement. The process is driven by asking the right questions by analyzing the ROM diagram of the design problem statement. The answers to the questions follow the lifecycle analysis of the product and its environment components. This paper completes a detailed procedure for Environment Analysis, which is the one of the three activities in Environment-Based Design (EBD).

Keywords: Design methodology, Conceptual design, Right questioning strategy, Question answering template, Environment-Based Design (EBD), Environment analysis, Environment structure

1. Introduction

1.1. Design methodology: looking at things in a sequence for design

Each and every one of design methodologies indeed presents a procedure that specifies a sequence of things that one should look at, starting with a description of a design problem and ending with a description of design solutions (Zeng & Gu, 1999). In the past five decades, a wide range of design methodologies have been proposed to tackle design problems. A product, as a solution for a design problem, is defined as a system connected to its environment by means of inputs and outputs (Hubka & Eder, 1988). Considering a design problem as a statement or a set of statements about a product-environment system, which is composed of a product and its environments around it (Zeng, 2004), design methodologies can be classified based on respective parts of the product-environment system that are focused on by a concerned methodology. Product-based design methodologies such as Morphology based design (Zwicky, 1969), TRIZ (Altshuller, 1984; Bacciotti et al., 2016; Borgianni & Matt, 2016; Cascini, 2012; Savransky, 2000), Quality Function Deployment (QFD) (Akao, 1972; Chan & Wu, 2002), Axiomatic Design (AD) (Suh, 1990) and Decision-Based Design (Chen et al., 2012; Hazelrigg, 2008; Ming et al., 2020) are the paradigm that considers a product as its focus, where one must have a product in the first place. Function-based design

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methodologies, including systematic design (Hubka & Eder, 1988; Pahl et al., 2007), Function-Behaviour-Structure (Gero, 1990), Function-Behaviour-States (Umeda et al. 1990), Structure-Behaviour-Function (Goel et al., 2009), and ontological basis of functions (Hirtz et al., 2002), assumes the existence of a functional structure for the design problem, from which design solutions can be usually abducted (March, 1976; Roozenburg, 1993). Recent efforts have been made to apply function based design models to engineering practices through physics-based reasoning and functional design framework (Liu & Lu, 2020; Medyna et al., 2012; Mokhtarian et al., 2017). The functions, behaviors, or physics are general relations from environment-product interactions to product-environment interactions at different levels of abstractions (Zeng & Gu, 1999). The affordance based design methodology is focused on interactions between a product and its environment, rather than the relations between the interactions (Maier & Fadel, 2009). User-Centred Design (UCD) (Norman, 1988), Environment-Based Design (EBD) (Zeng, 2004, 2011), and Situated Function-Behaviour-State (S-FBS) (Gero & Kannengiesser, 2006) are the third paradigm, which starts a design process from the environment (user in case of UCD, all environment in case of EBD, and context in case of S-FBS) of a product. This third paradigm is examined in John Dewey's logic framework in the context of situatedness and recursivity (Dixon & French, 2020). Particularly, user emotion and cognition, as an important part of environment, has attracted a great deal attentions in the design research community (Balters & Steinert, 2017; Gero & Milovanovic, 2020; Zhao et al., 2020). While this section lists only a small fraction of existing design methodologies, a comprehensive review has been conducted to analyze in details the design methodologies in the literature (Wynn & Clarkson, 2018).

1.2. EBD: design from the environment, for the environment and to the environment

Over the last 15 years, the author has always started with the following conceptual design problem in his design methodology course:

"Design a vacation house that can easily fly from one location to another."

Not surprisingly, almost all of the student designs would always start with the function "fly" or the product "house", which are critical parts of solutions to the problem. The solutions are mostly extended from a concept of flying such as balloon, helicopter propeller, and rocket propulsion. Those kinds of concepts produced in the conceptual design stage will commit a majority of costs in the later stage of designs. The variety of concepts indeed leads to different products in the aerospace industry.

However, a study conducted by McLellan and Nicholl (McLellan & Nicholl, 2011) indicates that when one were to design the concept of a product, the last thing to look at should be the product to be designed. Looking at the product first would generate fixation in the designer's mind, which leads to a wrong path and sequence of actions for a creative design (Chrysikou and Weisberg, 2005; Crilly, 2015; Crilly & Cardoso, 2017; Nguyen & Zeng, 2017; Viswanathan et al., 2014). Therefore, one should look at the "house" last in completing the design problem above.

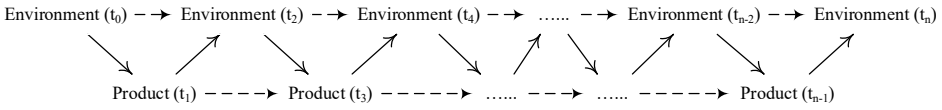


Figure 1. Design process: from unknown to known

Naturally, it is easier and more logical for human to start an action with something known. In the beginning of any design task, the only thing that might be known to designers is the existing environment whereas the product is unknown. The environment of a product is everything except the product itself (Zeng, 2001, 2002, 2004; Zeng & Cheng, 1991). The design process, regardless of how it is going in designers' minds, can be described in Figure 1, which shows that a product is designed to serve its environment, while simultaneously changing the environment due to its new presence in the environment. Therefore, in the case of designing a flying house, one should first look at its locations and users, among many other things. In general, the environment of a product to be designed should be the first thing to look at in a conceptual design.

Corresponding to the observation shown in Figure 1, Environment-Based Design (EBD) is a methodology that is focused on the environment throughout the entire design process (Zeng, 2004). In the Environment-Based Design (EBD), anything existing, known, and defined at the moment of designing can be viewed as environment whereas the unknowns to be created are the product. The unknowns, once designed or created in any form – abstract or concrete, will become known, which serves and changes the original environment. A product comes from its environment, serves for its environment, and brings changes to its environment (Zeng, 2011). Design is a process to generate a product that must fulfil human needs and that must survive in its inseparable environment by obeying the laws and regulations in the environment (Zeng & Cheng, 1991). Environment-Based Design embodies an environment evolution process, as shown in Figure 1, through its three activities – environment analysis, conflict identification, and solution generation in Figure 2.

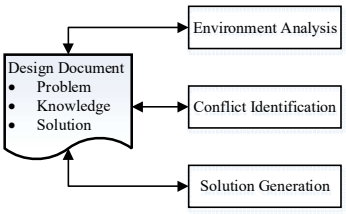


Figure 2. Environment-Based Design (EBD) (Zeng, 2011)

1.3. Contributions, objective and organization of this paper

The mathematical, linguistic, and cognitive foundations of EBD were reviewed in (Zeng, 2011). Like other design theories, which may suffer from instabilities of definitions of concepts and processes over their development (Dorst & Vermaas, 2005), the development of EBD and its applications have also experienced a co-evolutionary process through which the methodology has been subject to a continuous improvement. This present paper, followed by two other papers addressing different parts of EBD, aims to present a converged and stabilized form of EBD. This convergence has been made through its teaching to students with different disciplinary and cultural backgrounds and through its applications to industrial projects of radically different fields over the past 15 years. The major changes to be presented in this paper are listed as follows:

- Recursive Object Model (ROM) (Zeng, 2008) is now constructed as an instance of Axiomatic Theory of Design Modeling (ATDM) (Zeng, 2002), where three ATDM operations are defined: connection operation, constraint operation and interaction operation;
- Predicate relation in ROM (Zeng, 2008) is now decomposed into subject-verb and verb/proposition-object relations;
- A ROM matrix is now introduced for semantic computation of text;
- ROM based question asking (Zeng & Wang, 2009) is now related to a newly defined criteria for the rightness of questions. The question asking algorithm is not only for requirement elicitation but also for many other situations;
- Environment structure is now defined in a three dimensional space: lifecycle events, lifecycle time, and environment types; and
- Domain specific question asking (Zeng & Wang, 2009) is replaced by a question answering template.

The objective of all of the changes above is to systematically investigate the structure and analysis of environment by addressing the following three questions:

- Q1: Why is environment important?
- Q2: What is environment? and
- Q3: How to analyse environment?

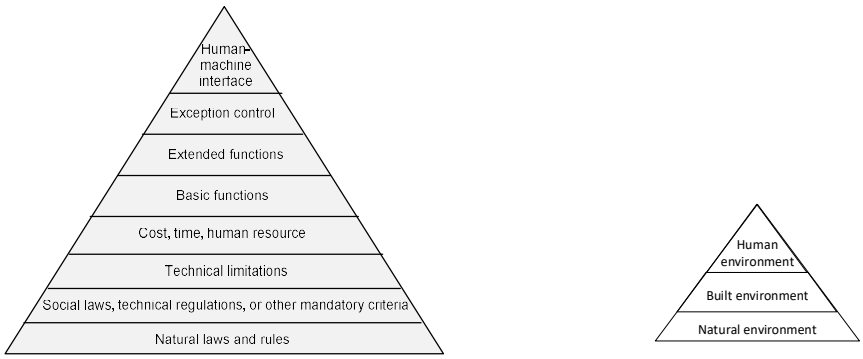
As Q1 was addressed in Section 1.2, the rest of the paper is organized as follows: Section 2, which answers Q2, presents the structure of an environment, which embodies a product that is realized. The product will become a part of the environment. Section 3, answering Q3, updates the environment analysis procedures by firstly defining the criteria for information quality in design, followed by a question asking algorithm based on ROM matrix, and finally with a question answering template based on the environment structure defined in Section 2. Section 4 briefly discusses the implication of the analysis on the modeling of design evolution and alternative approach (without ROM) to the environment analysis. Section 5 concludes the paper.

2. Structure of Environment

2.1. Environment: definition

2.1.1. Environment by types

The organization of environment components varies with different perspectives. Merriam-Webster defines environment as the circumstances, objects, or conditions by which one is surrounded[1]. Zeng and Cheng defines environment as working conditions in which an artifact lies in, including natural laws and actions (Zeng & Cheng, 1991). Chen and Zeng categorized product requirements, which are requirements on interactions between environment and product, into eight levels: natural laws, social law and regulations, technical limitation, cost, time and human resource, basic functions, extended functions, exception control level, and human-machine interface, as is shown in Figure 3 (Chen & Zeng, 2006). Obviously, different products emphasize different aspects of the requirements. For example, a software system is focused less on natural environment whereas mechanical parts may consider the natural laws and rules as the premium constraints of the design.



a) Eight levels of requirements (Chen & Zeng, 2006) b) Three basic types of environment

Figure 3. Types of environment related to product requirements

In the pyramid model shown in Figure 3a), those requirements at the lower levels are of more fundamental impact on developing a design solution in that they are more toward being mandatory. They can be divided into three major groups: constraining, functional, and usability requirements. The lower four are usually constraining requirements from natural and built environments since a product cannot survive without obeying them. Basic and extended functions are what a product is designed and meant for. The upper two aims to make the product user-friendly.

[1] https://www.merriam-webster.com/dictionary/environment, retrieved on September 6, 2020.

Among all those requirements, the lowest level of product requirements comes from the natural environment; the highest two levels of product requirements with some of the functional requirements serve for the purposes of human environment of the product; and the rest is all related to the built environment, which is made by humans. Correspondingly, we have three types of environments: natural, built, and human, as is shown in Figure 3b) (Zeng, 2004).

2.1.2. Environment by lifecycle

Product environment can also be defined in terms of the product lifecycle (Chen & Zeng, 2006). Different kinds of products may have different life cycles. For example, for a mechanical product, the life cycle can be divided into seven kinds of events, which are design, manufacture, sales, transportation, use, maintenance, and recycle, as shown in Figure 4 (Chen & Zeng, 2006). Multiple events in the lifecycle may happen currently or in sequence. For example, for the "fly" performance of a house, which is implied in the statement "the house can fly", the performance "fly" includes "take off", "move in the air" and "land" activities. Figure 5 shows the composition of product environment in terms of its type and lifecycle. This composition provides a guidance for a designer to have a comprehensive perception of the environment in which the designed product will work.

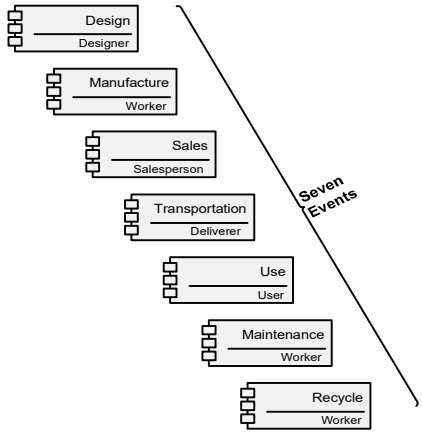


Figure 4. Seven events in an example product life cycle (Chen & Zeng, 2006)

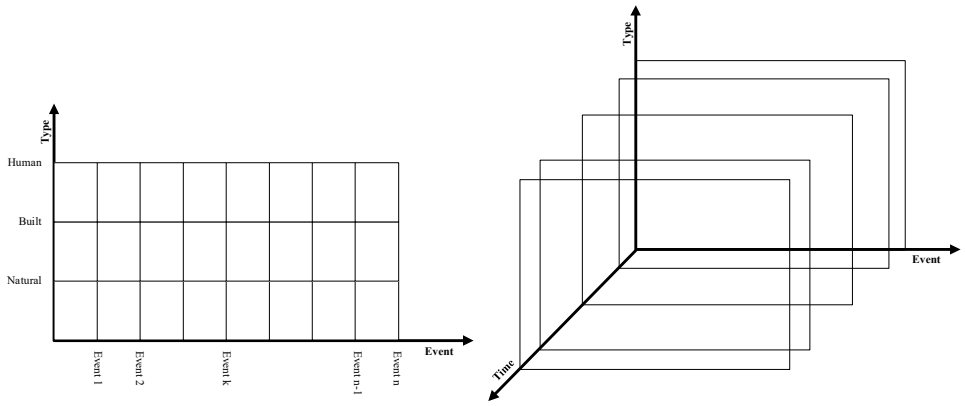


Figure 5. Composition of product environment in terms of its types and life cycle (time and events)

2.2. Environment: recursive structure

Simon defined design as an activity to change a situation to a desired one (Simon, 2019). Ultimately, design in general and creative design in particular is to generate something new from nothing, which brings

changes to its current situation (Zeng, 2011). Hence, design in nature is a simultaneous evolution of design problem, design knowledge and design solution (Dorst & Cross, 2001; Maher & Tang, 2003; Zeng & Cheng, 1991; Zeng & Gu, 1999), for which the major challenge in its formal studies is how to represent something unknown and to be defined/refined in the future. Due to this co-evolutionary nature of the conceptual design process and the composition of product environment illustrated in Figure 5, a good conceptual design representation must satisfy the following criteria (Zeng & Gu, 1999; Zeng, 2002):

- Be able to represent design problem, design knowledge, and design solutions in one framework;
- Be able to represent the unknown and probably novel structure of design problem, design knowledge, and design solutions that are to be evolved from the present state; and
- Be able to represent the above in terms of product environment.

In fulfilling the criteria above, the axiomatic theory of design modeling (ATDM) (Zeng, 2002) can be used as follows

$$\oplus E = E \cup (E \otimes E), \tag{1}$$

where $\oplus E$ is the structure of Environment E.

In the context of an application, a group of primitive environment components can always be defined as the components that cannot or need not be further decomposed (Zeng, 2008; Zeng, 2002):

$$\oplus E_i^a = E_i^a, \exists n, i = 1, \dots, n, \tag{2}$$

where E_i^a is a primitive environment component and n is the number of the primitives. These primitive environment components form an algebraic basis to represent the environment evolution (which is design). Examples of primitive environment components include natural, built, and human environments. The composition shown in Figure 5 can be used to define the primitives of a design. The selection of primitives is subjective, depending on designer's experience and expertise (Zeng & Cheng, 1991; Zeng & Gu, 1999).

Table 1. Elements of Recursive Object Model (ROM), adapted from (Zeng, 2008)

Type			Graphic Representation	Definition
Object	Primitive objects		E_i^a	Everything in the universe is an object.
	Object structure		$\oplus E$	It is an object that includes at least two other objects in it.
Relations	Connection		E_i --- l ---▸ E_j	It is to connect two objects that do not constrain each other.
	Constraint		A ●▸ E_i	It is a descriptive, limiting, or particularizing relation of one object to another.
	Predicate	Subject-verb	S_i ▪— v	It describes an act of an object on another or that describes the states of an object.
		Verb/proposition-object	v/p ▸ O_j	

Mathematically, the ATDM provides a means to represent the structure of an artefact being designed during the design process, which puts the knowns and the unknowns into the same representation framework. This is critical for creative design in that the structures of design solutions are often new and emergent; thus, it cannot be represented in a known structure. Derived from the ATDM, Recursive Object Model (ROM) (Zeng, 2008) is a conceptual tool to represent objects and their relationships for the meaning structure, which can be used to represent and reason about design activities. The ROM has two kinds of objects and three kinds of relations, as is shown in Table 1.

Mathematically, relations in ROM can be defined as connection operation, constraint operation, and interaction operation.

Connection operation, denoted by Δ, is a relation between two objects (O_1 and O_2) that will generate a new object (O_3) without changing the properties of the objects (O_1 and O_2).

$$O_3 = O_1 \Delta O_2,\ \exists O_1, O_2. \tag{3}$$

Connection relation includes spatial connections (such as ∩ (intersection), / (difference)) and logical connections (such as ¬ (negation), ∧ (conjunction) and ∨ (disjunction)).

Constraint operation, denoted by *, is a relation from a constraining object (O_1) to a constrained object (O_2), which will generate a new object (O_3) by defining a property O_1 of the object O_2.

$$O_3 = O_1 * O_2. \tag{4}$$

Interaction operation, denoted by ⊗, is a relation from one object (O_1) to another object (O_2) that will generate a new object O_3 that are defined by properties different from those defining O_1 and O_2.

$$O_3 = O_1 \otimes O_2 \xleftrightarrow{\Delta} [\exists O_1, O_2, O_3, \rho, (O_3 = \rho(O_1, O_2)) \vee (O_3 = \rho(O_2, O_1))]. \tag{5}$$

While connection and constraint operations are quite straight-forward in a natural language, there are four kinds of basic interactions identified in a ROM diagram, corresponding to natural languages, as shown in Table 2. Figure 6 gives ROM diagrams of examples in English, French and Chinese.

Table 2. Four types of interactions Argument价：名词关联数

Type	Sentence pattern	Example
Type 1	Subject + Action Verb + Object	The factory makes tools.
Type 2	Subject + Action Verb + Direct Object + Preposition + Indirect	The factory makes tools for us.
Type 3	Object Subject + Action Verb + Direct Object + Indirect Object	The factory makes us tools.
Type 4	Subject + Action Verb	The factory succeeds.

3. Environment Analysis

The greatest challenge in conceptual design and strategic decision making is the uncertainty implied in problem statement (Zeng, 2016). This uncertainty may originate with two sources: first, the problem statement could include unnecessary product information, which is given only because some functions carried by the design decisions are desired. Second, lack of information in the problem statement keeps the problem open-ended, which leads to multiple options for the design decision making. Therefore, it is critical to determine decision directions to effectively and efficiently deal with an open-ended conceptual design and strategic decision making problem.

Environment analysis, as the first activity in the EBD, aims to identify environment components and their relationships that are critical to the product to be designed. Starting from a design problem statement, the environment analysis identifies the product to be designed, the *explicit* and *implicit* components of product environment, and the mutual interactions between the product and the environment components and between the environment components.

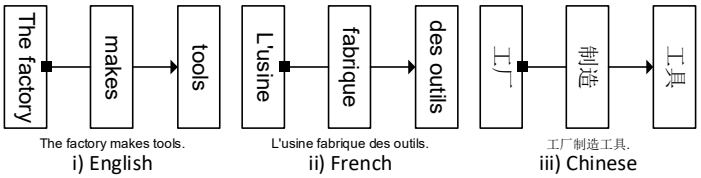
A good environment analysis finds high quality information that is the premise for an effective and efficient conceptual design, which should follow the two criteria below:

1) The environment analysis should identify the environment components and their relationships that are necessary for a given design stage; and
2) The environment analysis should identify all of the environment components and their relationships that are definable at a given design stage.

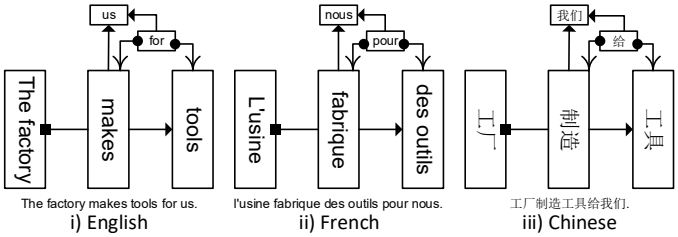
These two criteria intend to avoid the under- or over- definition of a design problem at any stage of the design. The three steps below aim to satisfy the two criteria above:

- **Determine decision directions by asking design questions based on a design problem statement:** As design is a recursive process (Zeng & Cheng, 1991), one cannot generate meaningful design solutions unless one has a meaningful understanding of the design problem; one cannot develop a meaningful understanding of the design problem unless one has a concept of design solutions in mind.

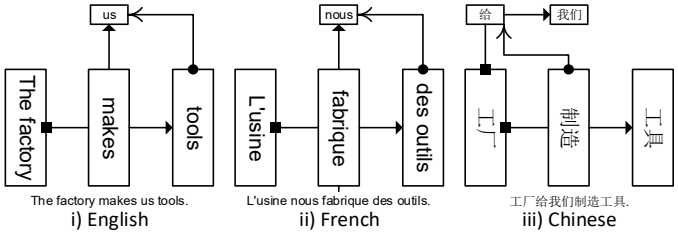
This recursive interdependence between a design problem and design solutions calls for good decision directions in fulfilling the requirements specified in the design problem statement. A question strategy will be introduced in this section to support an efficient and effective determination of a good decision direction based on a design problem statement.



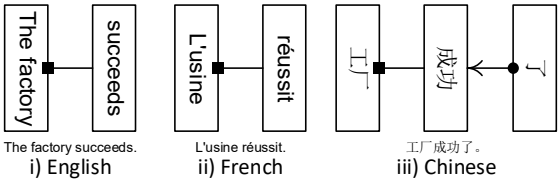
a) Type 1 interaction: subject-verb-object



b) Type 2 interaction: subject-verb-object 1-proposition-object 2



c) Type 3 interaction: subject-verb-object 1-object 2



a) Type 4 interaction: subject-verb

Figure 6. Basic interactions in natural languages: ROM representation

- **Clarify the design problem statement by answering questions following the decision direction:** Since product environment is the source of product requirements, identification of both explicit and implicit product environment components is critical to find product requirements that can be sufficiently defined at the ongoing design stage. In this section, a roadmap and templates are provided to guide designers in answering design questions.
- **Formulate an environment system from a design problem statement:** As design is a co-evolutionary process (Dorst & Cross, 2001), a product will become a part of the environment once it is generated. An environment system is thus implied in a design problem statement at any design stage. The environment system embodies all of the information available at a design stage. Through this

activity, product and environment components as well as their mutual relationships will be recognized and formulated into an environment system.

In the following subsections, the details of these three activities will be introduced with the support of an example to design a flying house as follows:

Design a vacation house that can easily fly from one location to another location.

3.1. Questioning strategy to determine decision direction

As was argued by Eris (Eris, 2004), an effective decision making process cannot be performed without having high quality information; and high quality information cannot be acquired and generated without performing an effective inquiry process. The inquiries are achieved by asking 5W1H questions (what, who, when, where, why and how), which can be done based on one's expertise and experience.

However, expert designers could tend to ask questions biased toward their expertise due to fixations whereas novice designers could struggle to ask meaningful questions. Experiments show that people tend to first ask questions related to the product (McLellan & Nicholl, 2011). Wang and Zeng has proposed a question strategy to ask the right 5W1H questions for eliciting product requirements (Zeng & Wang 2009), which is naturally focused on the environment of the product to be designed. A right question is the one asked in the right time about the right object with the right type and to the right resources. An algorithmic implementation of the question strategy, which is illustrated in Figure 7, is now proposed to cover the entire process of design and for a broader range of problems.

The question strategy includes two main parts: 1) to determine the sequence of the object to be asked (when to ask about which object); and 2) to decide what type of questions to be asked. The first part will produce an object-list according to which questions will be produced, based on a set of questioning rules whereas the second part will generate questions for a selected object based on a question template. ROM analysis will be conducted before these two parts are applied following the rules and definitions provided by Zeng (Zeng 2008).

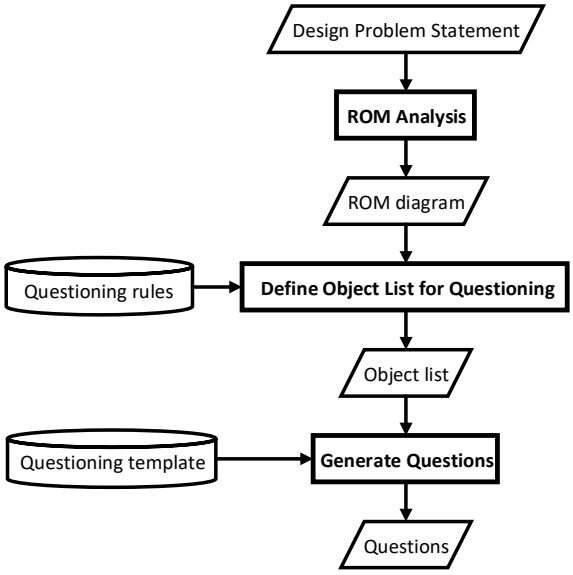


Figure 7. Question generation process in Environment Analysis

3.1.1. ROM analysis

The first step in determining the decision direction is to understand the design problem statement. Though this understanding can be achieved through one's personal capability and experience, the ROM analysis process can be used to achieve a more objective understanding by transforming the original design

problem statement described by natural language into a ROM diagram. ROM analysis is methodology oriented while experts are mostly experience based.

Figure 8 is the ROM diagram corresponding to the flying house design problem. It shows the components included in the initial environment system and the relations between the components.

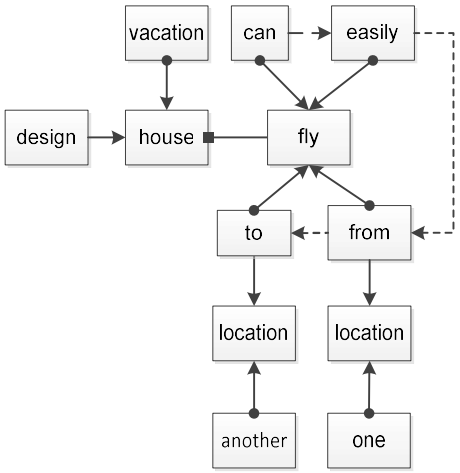


Figure 8. ROM analysis of problem statement: "Design a vacation house that can fly easily from one location to another."

3.1.2. Object list for questioning

The second step in determining the decision direction is to identify the objects that need to be questioned and the sequence to question them by using the design problem statement ROM diagram as the input. This is completed in four steps as following:

1) Code objects in the ROM diagram
2) Construct ROM matrix
3) Rank ROM objects, and
4) Define object list for questioning.

In this subsection, the details of the four steps will be introduced, respectively.

1) Code objects in the ROM diagram

Every object in the ROM diagram is assigned a number in a continuous manner. Corresponding to the ROM diagram in Figure 8, the coded ROM diagram is shown in Figure 9.

2) Construct ROM matrix

A coded ROM diagram can be represented in a matrix form by assigning relations according to the following equation:

$$r_{ij} = \begin{cases} 1 & objects\ i\ and\ j\ have\ a\ connection\ relation \\ 2 & there\ is\ a\ constraint\ relation\ from\ object\ i\ to\ object\ j \\ 3 & objects\ i\ and\ j\ have\ a\ subject-verb\ relation \\ -3 & objects\ j\ and\ i\ have\ a\ subject-verb\ relation \\ 4 & objects\ i\ and\ j\ have\ a\ verb/proposition-object\ relation \\ -4 & objects\ j\ and\ i\ have\ a\ verb/proposition-object\ relation \\ 0 & otherwise \end{cases} \tag{6}$$

Please note that the values assigned to each relation r_{ij} do not have any implication at this specific stage. They only distinguish different kinds of relations. The relations for coded ROM diagram in Figure 9 can be shown in Table 3. By default, the number for a relation is 0.

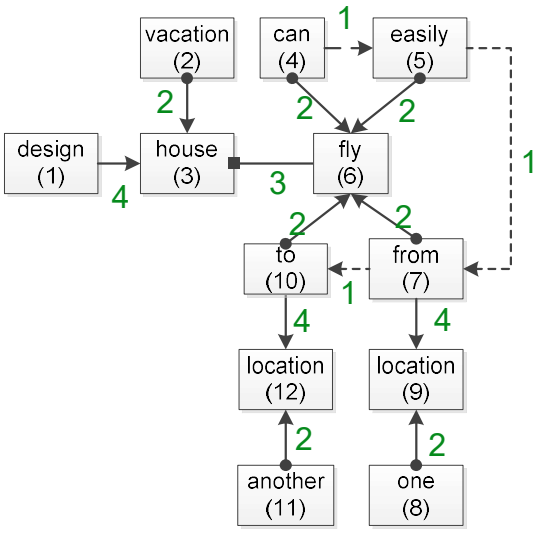


Figure 9. Coding ROM objects in the problem statement: "Design a vacation house that can fly easily from one location to another."

Table 3. Relations for the coded ROM diagram in Figure 9: Row = Object j, Column = Object i

Object	1	2	3	4	5	6	7	8	9	10	11	12
1	0	0	4	0	0	0	0	0	0	0	0	0
2	0	0	2	0	0	0	0	0	0	0	0	0
3	-4	0	0	0	0	3	0	0	0	0	0	0
4	0	0	0	0	1	2	0	0	0	0	0	0
5	0	0	0	0	0	2	1	0	0	0	0	0
6	0	0	-3	0	0	0	0	0	0	0	0	0
7	0	0	0	0	0	2	0	0	4	1	0	0
8	0	0	0	0	0	0	0	0	2	0	0	0
9	0	0	0	0	0	0	-4	0	0	0	0	0
10	0	0	0	0	0	2	0	0	0	0	0	4
11	0	0	0	0	0	0	0	0	0	0	0	2
12	0	0	0	0	0	0	0	0	0	-4	0	0

3) Rank ROM objects according to the number of constraint and predicate relations

Table 4 shows the number of constraint and predicate relations on the objects in the ROM diagram given in Figure 9. It can be easily seen that object 6 is the central object in this text since it has the greatest number of constraint and predicate relations with other objects.

Table 4. Number of constraint and predicate relations on an object

Number of relations	5	3	2	1
Object	6	3	7, 10, 9, 12	1

Figure 10 illustrates the structure of the objects around the central object in terms of constraint and predicate relations derived from the ROM diagram in Figure 9.

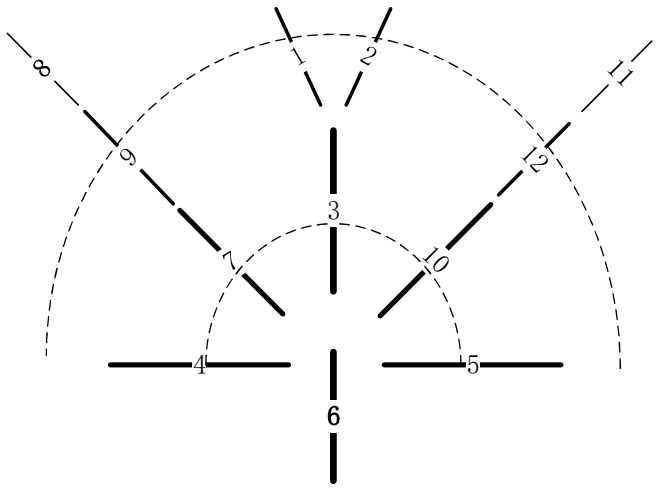


Figure 10. Structure tree of constraint and predicate relations of objects

4) Define object list for questioning

An object list will determine when to ask questions about which object. Using the information in Table 3 and Table 4, rules in Table 5 can be applied to find the order of questioning objects for design-related tasks. Rules can be different for different applications.

Table 5. Rules for finding objects to question

Rule #	Rule description
Rule 1	Among all of the candidate objects, the object with the most undefined constraints and predicates should be considered for questioning first.
Rule 2	Before an object can be asked a question, the objects constraining or predicating them should be asked.

According to the rules in Table 5, central objects, which have the most number of relations going toward them, will be questioned last. The objects that constrain an object may refine the meaning of the object; thus, the meaning of these constraining objects shall be first clarified by being questioned.

The tree structure shown in Figure 10 indicates that the constraints and predicates around the central object could be categorized in two ways: in depth for which objects are grouped along each branch of the tree; in width for which objects are arranged in the same layer away from the central object. Therefore, the questions can be generated on the applicable objects and object group following three schemes: depth first, width first and hybrid (Wang et al., 2015; Zhao & Zeng, 2019).

For the flying house design problem statement given earlier, an object list for questioning is given in Table 6, which follows the width first scheme.

Table 6. Object list to be questioned

Number of relations	5	3	2	1
Object	6	3	9, 12	1, 7, 10
Questioning object list	**1, 2, (8, 9), (11, 12), 3, 6**			

The process generating the questioning object list is given in Table 7.

Table 7. Process to generate questioning object list for the ROM diagram in Figure 9

Rules from Table 5	Conditions	Objects that should be considered for questioning
Rule 1 to the ROM	Object **6** has the most undefined constraints and predicates	**6**
Rule 2 to Object **6**	Object **3** predicates Object **6** and Object **4**, **5**, **7**, and **10** constrain Object **6**	**3, 4, 5, 7, 10**
Rule 1 to Object **3**, **4**, **5**, **7**, and **10**	Object **4** and **5** are self-defined while Object **3** has 2 predicate relations and 1 constraint relation, Object **7** and **10** have one predicate relation.	**3 followed by 7 and 10**
Rule 2 to Object **3**	Object **1** predicates Object **3** and Object **2** constrains Object **3**.	**1 or 2**
Rule 1 to Object **1** and **2**	No other constraint or predicate relations on Object **1** or **2**.	**Question 1 and 2**
Rule 2 to Object **7 or 10**	Object **9** predicates Object **7** and Object **12** predicates Object **10.**	**9 or 12**
Rule 1 to Object **9** and **12**	Object **8** predicates Object **9** and Object **11** predicates Object **12.**	**8 or 11**
Rule 2 to Object 8 and 11	No other constraint or predicate relations on Object **8** and **11**; Object **8** and **11** are self-evident.	**Question (11, 12) and (8,9)**
Rule 1 to the ROM	Object 1, 2, 4, 5, 7, 8, 9, 10, 11 and 12 are either questioned or self-evident	**Question 3 and 6**

3.1.3. Generation of questions

The third step in determining the decision direction is to generate questions for the object identified in the previous step. A sample question template is given in Table 8 to generate questions for an identified object. It must be noted that this template is by no means complete and the readers can develop their own refined ones based on a comprehensive experiment and/or on different purposes.

Table 8. Question template for object analysis

#	Conditions	Question template
T1	For a concrete, proper, or abstract noun object N without any constraint	What/Who is N?
T2	For a concrete, proper, or abstract noun N with an adjective constraint A	What is A N?
T3	For an noun object A constraining an noun object N	What is A? What is/are A N?
T4	For a verb V with its subject N1 and object N2	What do you mean by V in the statement "N1 V N2"? How do/does N1 V N2? Why do/does N1 V N2? When do/does N1 V N2? Where do/does N1 V N2?
T5	For a verb object V constrained by an adverb A with its subject N1 and object N2	What do you mean by V A? Why do/does N1 V A N2? When do/does N1 V A N2? Where do/does N1 V A N2?
T6	For a verb V with an object N, but missing its subject	What/Who V N?

Following the question template above, questions are generated according to the object list in Table 6 as shown in Table 9.

Table 9. Questions to clarify the problems for flying house design

Object	Question Template and Conditions	Questions
1	"Design" is a verb, T4, T5 and T6 is applied.	Q1: Why design a vacation house that can fly easily from one location to another? Q2: Who will design the house? Q3: How to design the house? Q4: When to design the house? Q5: Where to design the house?
2	"Vacation" is a concrete noun, T1 is applied.	Q6: What is "vacation"?
(8, 9)	"Location" is constrained by "one", T2/T3 is applied.	Q7: What is "one location"?
(11, 12)	"Location" is constrained by "another", T2/T3 is applied.	Q8: What is "another location"?
3	"House" is a concrete noun, T1 is applied.	Q9: What is a "house"?
	"House" is constrained by "vacation", T2/T3 is applied.	Q10: What is a "vacation house"?
6	"Fly" is a verb, T4, T5 and T6 can be applied.	Q11: What is the meaning of "fly easily"? Q12: How can the "house fly easily from one location to another"? Q13: When does the "house fly easily from one location to another"?

After the questions are asked, they can be answered based on the designer's knowledge and experience. The answers will differentiate from one designer to another; still, Section 3.2.1 will provide a guideline for effective design question answering to enhance the chances to reach a more innovative solution with a faster convergence rate.

3.2. Template for answering design related questions

The answers to the design questions generated in Section 3.1 must serve the objectives of environment analysis, which are to identify the product to be designed, the explicit and implicit components of product environment, and the mutual interactions between the product and the environment components and between the environment components. Therefore, this section will first show a question-answering guideline following the composition of product environment given in Figure 5.

Table 10. Guideline to answer question in Table 8

#	Questions	Guideline
G1	What/Who is N? N: a concrete, proper, or abstract noun object What is A N? A: an adjective constraint	a) If (A)N is the product to be designed, then the answer should address 1) the purpose of (A)N; 2) the definition of (A)N according to the life cycle of the product and the categorization of environment components for each event in the lifecycle; b) Else, if N is an environment component of a product, then the answer should define (A)N according to the life cycle of the environment component and the categorization of the other environment components for each event in the lifecycle; c) Else, the components and attributes of N should be described.
G2	What/Who do/does V N? V: a verb	For N1 that V N, the answer should define the components and attributes of N1 in the context of V N.
G3	When do/does N1 V N2? When do/does N1 V A N2?	The answer may assume one of the following two forms: a) In/On a time, N1 V(A) N2; b) When/During/While N3 V_a N4, N1 V(A) N2.
G4	Where do/does N1 V N2? Where do/does N1 V A N2?	The answer may assume one of the following two forms: a) In/Along/Through a place, N1 V(A) N2; b) N3 V_a N4, where N1 V(A) N2.
G5	Why do/does N1 V N2? Why do/does N1 V A N2?	The answer should be organized as: To V_a N_a, N1 V (A) N2.
G6	What do you mean by V? What do you mean by V A? How do/does N1 V N2?	a) If the subject (N1) or object (N2) of V is not the product, then the answer should include all activities included in V-ing in the context of N1 and N2; b) Else, skip the question and leave for solution generation.

3.2.1. Question-answering guideline

Answers to design-related questions must serve the purpose of environment analysis by providing the information sufficiently and necessarily available at the moment that can assist in the identification and definition of the explicit and implicit components of product environment, and the mutual interactions between the product and the environment components and between the environment components. By sufficiently available we mean that all of the information that can be well defined will be given in the answers whereas by necessarily available we mean that all of the information that is included in the answers comes only from a deductive process based on available information and resources. A guideline is provided in Table 10 to guide the answering of the questions listed in Table 8.

Following the guideline given in Table 10, answers to questions in Table 9 are listed in Table 11, which will be added into the ROM diagram in Figure 8 one by one.

Table 11. Answers for questions in Table 9

Q1: Why design a vacation house that can fly easily from one location to another?
Q2: Who will design the house?
Q3: How to design the house?
To provide a place for people to stay in and do everyday activities during a vacation, designers design a vacation house using a methodology. The house transports people easily during their vacation.
Q4: When to design the house?
Q5: Where to design the house?
(These two questions can be skipped for now. Their significance can be seen in the conflict identification stage. The impact of answering or skipping a question will be demonstrated in the solution generation stage.)
Q6: What is “vacation”?
A period of time during which people leave home for rest and recreation. A vacation includes the following activities: the people plan the vacation; the people travel to the destination; and the people enjoy the time in the destination.
Q7: What is "one location"?
Q8: What is "another location"?
There are three kinds of locations: starting (one) location, intermediate location, and destination (another) location. The starting and the destination locations include home, beaches, lakes, forests, mountains, and cities. The intermediate location is the space in the air.
Q9: What is a “house”?
It is where people do daily activities such as cooking, dinning, cleaning, relaxing, sleeping, and entertaining. The lifecycle of a house includes design, manufacturing, service, maintenance, and demotion. In the current design, we will consider only conceptual design (design), transporting people, and serving daily activities. For the event of conceptual design, natural environment can be ignored while the human environment is designer and the built environment is technical resources and design tools. For the event of transporting people, the natural environment include earth, air, sky, and all of the natural locations in which the house will stay as well as animals and birds in the environment; the built environment includes all of the vehicles along the path of transportation, all of the man-made places in which the house will stay; the human environment includes the vacationers and other people to appear in the environment. For the event of serving daily activities, we can look for environment components for activities such as cooking, dinning, cleaning, relaxing, sleeping, and entertaining, which were identified in the beginning of this answer.
Q10: What is a “vacation house”?
It is a house where people stay in and do everyday activities during a vacation.
Q11: What is the meaning of "fly easily”?
Q12: How does a “vacation house fly from one location to another location”?
(Since the subject of the verb “fly” is the product “house”, Q11 and Q12 are left for solution generation.)
Q13: When does a “vacation house fly from one location to another location”? G3b)
The house will fly when people go on vacation.

For the convenience of understanding, the design problem statement corresponding to the ROM diagram in Figure 13, can be updated as follows:

Designers will design, using a design methodology and technical resource, a vacation house that can easily transport people who would travel to various destinations. The destinations are starting locations, immediate locations and final locations, which include home, air, beaches, lakes, forests, cities, and mountains. In those destinations, there are birds, animals, people and various artefacts. People will plan their vacation during which they will leave their home for rest and recreation. The house provides a place for the people to stay and to enjoy their vacation time. In the house, people will conduct activities such as

cooking, dinning, cleaning, relaxing, sleeping, and entertaining. The vacation house will transport people, who will fly the house when they want to leave home for rest and recreation during a period of time.

If designers are satisfied with the information sought through the questioning-answering process, then the process can stop; otherwise, more questions can be generated based on the updated ROM diagram by repeating the process described above.

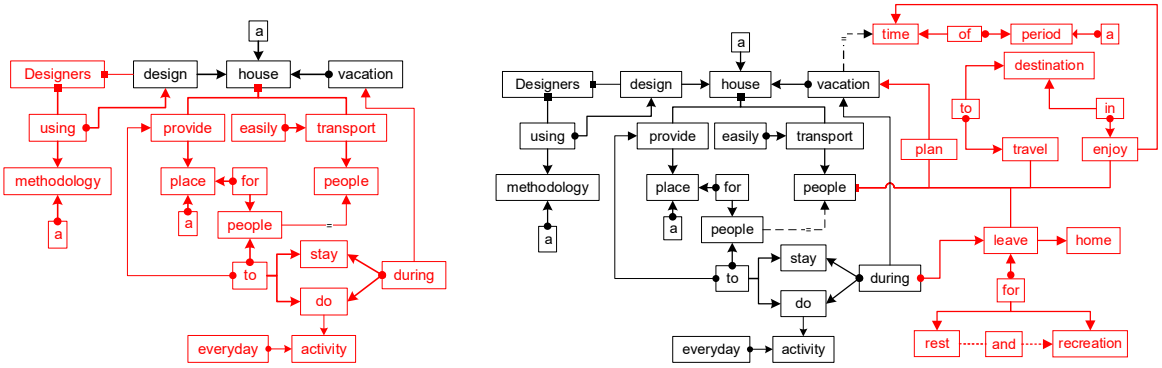


Figure 11. ROM diagrams updated by the answer of Q1-3 and Q4-7, respectively

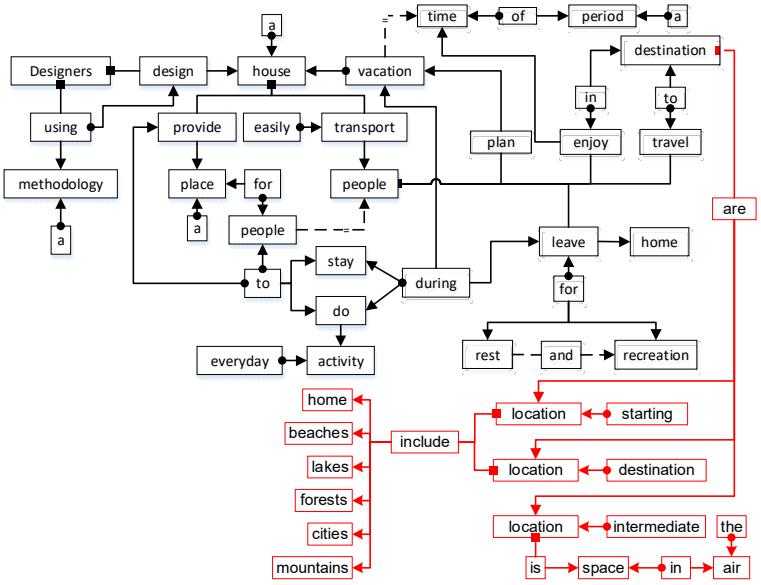


Figure 12. ROM diagram updated by the answer of Q8

4. Discussions

4.1. Analysis of information content in the result

The effect of EBD can be roughly observed by comparing the numbers of words used to describe product, environment, and their relationships included in the two ROM diagrams in Figure 8 and Figure 13. The following can be observed from Table 12:

- The original design of the "house" as presented in Figure 8 is now evolved into two design of both the "house" and the design project team "designers" as presented in Figure 13;
- The description describing the environment of "house" design has evolved from 5 words to 99 words and the relationships between the product "house" and its environment has increased from

4 words to 11words while the number of occurrences of the word "house" has increased only from 1 to 4 without any refinement. Therefore, the environment analysis is focused on the environment of the product;

- Two products "house" and "designers" are coupled in a sense that the project management depends on how the house is designed and the house depends on how the designers are selected and resources are assigned. Therefore, design is a process biased toward designers' capabilities.

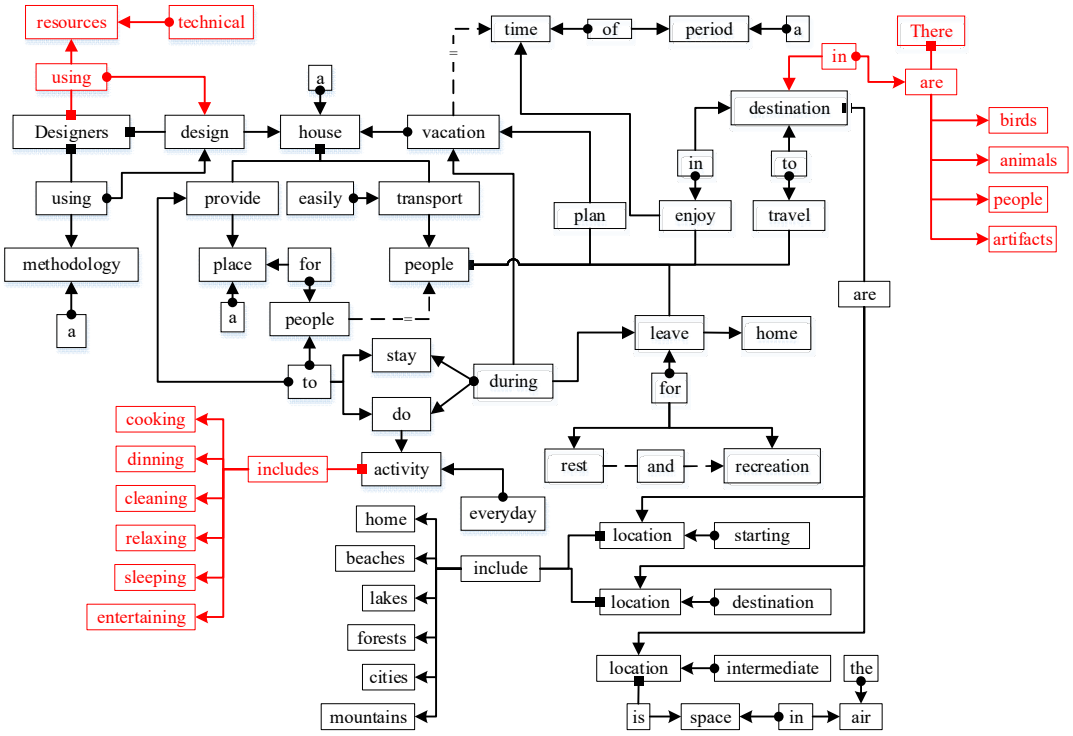


Figure 13. ROM diagram updated by the answer of Q9-13

4.2. Different strategies for question asking

There are three strategies in question-driven information search: depth first, width first, and hybrid first, corresponding to each of which designer's mental stress and performance can be different (Wang et al., 2015; Zhao & Zeng, 2019). The rules given in Table 5 are indeed a width first strategy, as can be seen from Table 9. Different rules can be configured for different purposes, which will in turn reformulate the design problem differently. The strategy will thus lead to different design solutions according to the conditions for design creativity listed in (Nguyen & Zeng, 2012; Zeng, 2001, 2004; Zeng et al., 2011): 1) formulating a design problem differently; 2) extending design knowledge; and 3) changing the environment decomposition.

4.3. Can Environment Analysis be done without ROM?

As can be seen from the paper, ROM diagram can be used to generate questions that are aligned with the philosophy of EBD. If one naturally designs with environment-centred thinking, they will be able to ask the same kind of questions following the same order as if they were using EBD. They can follow the guideline to answer questions. However, when the size of design text is large, it becomes infeasible for human designers to follow the same process, designer experience becomes critical to ask the right questions. The role of ROM analysis to complex design is like the role of mesh generation to complex finite element

analysis. Therefore, an effective and robust ROM analysis is indispensable to tackle complex design problems.

Table 12. Evolution of information content following EBD

ROM	Product	Environment	P-E Relationships
Figure 8	House (1 occurrence)	Vacation; One location; Another location. (5 words)	(House) fly easily from (one location) to (another location) (4 words)
Figure 13	House (4 occurrences)	Designers using a design methodology and technical resource; People who would travel to various destinations; The destinations are starting locations, immediate locations and final locations, which include home, air, beaches, lakes, forests, cities, and mountains; In those destinations, there are birds, animals, people and various artefacts; People will plan their vacation during which they will leave their home for rest and recreation; A place for the people to stay and to enjoy their vacation time; Activities such as cooking, dinning, cleaning, relaxing, sleeping, and entertaining; People want to leave home for rest and recreation during a period of time. (99 words)	(Designers) will design (a vacation house); (The house) provides (a place) In (the house), (people) will conduct (activities) (The vacation house) will transport (people); (People) will fly (the house) when (they want to leave home); (11 words)
	Designers -project plan (1 occurrence)	Design methodology and technical resource; A vacation house that can easily transport people; People who would travel to various destinations; The destinations are starting locations, immediate locations and final locations, which include home, air, beaches, lakes, forests, cities, and mountains; In those destinations, there are birds, animals, people and various artefacts; People will plan their vacation during which they will leave their home for rest and recreation; The house provides a place for the people to stay and to enjoy their vacation time; In the house, people will conduct activities such as cooking, dinning, cleaning, relaxing, sleeping, and entertaining; The vacation house will transport people, who will fly the house when they want to leave home for rest and recreation during a period of time. (125 words)	(Designers) will design (a vacation house) (Designers) using (a design methodology and technical resource) (3 words)

5. Conclusions

This paper presents a thorough investigation into the concept of environment from the four aspects below:

1) Environment is the first thing that should be considered in an effective and efficient conceptual design process. Design comes from environment, serves the environment, and changes environment. This answers why environment is important for design;

2) Environment can be defined from three perspectives: type (natural, built, and human), activities (such as design, manufacturing, transportation, use, maintenance, recycle and disposal), and time. Environment structure is recursive and can be represented by Axiomatic Theory of Design Modeling (ATDM) in general and Recursive Object Model (ROM) in particular. This answers what is environment; and
3) Environment analysis, as one of the three activities in Environment-Based Design (EBD), aims to develop a necessary and sufficient understanding of the environment for a product to be designed. Environment analysis consists of three activities: 1) determine decision directions by asking design questions based on a design problem statement; 2) clarify the design problem statement by answering questions following the decision direction; and 3) formulate an environment system from a design problem statement. The right design questions are generated for the right object in the right sequence with the right question type by analyzing the ROM diagram corresponding to the design problem statement. The questions are answered following a guideline, which should go to a right source, aiming to provide necessary and sufficient information to define the environment of a desired product in the concerned design stage. The guideline is developed on the basis of a composition of product environment in terms of the definition of environment. This answers how to analyze environment.

This present paper also discussed the few aspects of environment analysis including its potentials in analysing product evolution, impact of different questioning strategies, and the implementation of the methodology without ROM. Other activities of EBD such as conflict identification and solution generation will be systematically introduced in separate papers.

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